

Troubleshooting Methodology — Guided Notes ANSWER

KEY

Unit 5.1.1 | CompTIA Network+ N10-009 Obj. 5.1 | N10-008 5.1

For instructor use / distribute after students complete the notes.

Question 1

List the CompTIA seven-step troubleshooting methodology in order.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

Answer: 1. Identify the problem. 2. Establish a theory of probable cause. 3. Test the theory to determine the cause. 4. Establish a plan of action. 5. Implement the solution or escalate. 6. Verify full system functionality and implement preventive measures. 7. Document findings, actions, and outcomes.

Question 2

Step 1 says to 'question the obvious.' Give an example of an obvious check that resolves a surprising number of network tickets — and explain why it's easy to skip.

Answer: Examples: Is the cable plugged in? Is the port administratively shut down? Is the device powered on? These are easy to skip because they feel beneath a trained technician — but they resolve a significant percentage of real tickets.

Question 3

Why is Step 3 (test the theory) separated from Step 5 (implement the solution)? What mistake happens when a technician combines testing and implementing?

Answer: Testing confirms the cause through observation without changing anything. If a technician changes the config to 'test' whether that fixes it, they may introduce a second problem, lose the ability to roll back, or fix a symptom without identifying the actual cause.

Question 4

Step 4 requires establishing a plan before implementing. A good plan includes: the action to take, the expected _____, what to check to confirm success, and a _____ plan if the change doesn't work.

Answer: Outcome/result; rollback.

Question 5

Bottom-up troubleshooting starts at Layer _____ and works upward. This approach guarantees you don't miss a lower-layer problem masquerading as a higher-layer issue. The tradeoff is _____ — it checks many layers before reaching the actual problem.

Answer: 1 (Physical); speed/efficiency.

Question 6

Divide-and-conquer troubleshooting typically starts at Layer _____. If the device can ping its _____ successfully, you know Layers 1-3 are likely working and you focus upward. If not, you focus downward.

Answer: 3 (Network); default gateway.

Question 7

Step 7 — documentation — is often skipped. Name two concrete negative outcomes that happen when documentation is absent from the troubleshooting process.

Answer: Examples: the next technician who encounters the same issue starts from scratch, wasting time; recurring problems are never identified as recurring because there is no history; institutional knowledge exists only in one person's memory and is lost when they leave.

Question 8 — Real World Application

REAL WORLD: A tech gets a ticket that second-floor users can't reach the internet. They assume it's the uplink cable (same problem last week), swap it, and leave. Fifteen minutes later, the ticket reopens. Identify which step they skipped and what the correct process would have revealed.

Answer: Skipped Step 1 (information gathering) and Step 2 (forming and ranking theories). Correct process: check the switch for error messages — in this case, the uplink port was err-disabled by a loop that appeared an hour earlier. The cable was never the problem.